

Enterprise **Active Directory** & SIEM Home Lab

VMware Workstation Pro Edition · Complete Step-by-Step Implementation Guide

Virtualization Platform	VMware Workstation Pro
Domain Controller	Windows Server 2025
Domain Client	Windows 10 / 11
Attack Machine	Kali Linux (Latest)
SIEM Solution	Splunk Enterprise + Universal Forwarder
Difficulty Level	Beginner — Intermediate

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1. Introduction & Lab Overview

This guide walks you through building a fully functional Active Directory security lab from scratch. You will stand up a real enterprise identity environment, generate authentic Windows Event Log telemetry, forward it to a Splunk SIEM, and simulate attacker activity from Kali Linux to observe detection in real time.

This guide uses **VMware Workstation Pro** as the hypervisor, which has been free for personal use since May 2024, and provides reliable snapshots, flexible virtual networking, and good performance for running four VMs concurrently.

What You Will Build

The lab consists of four virtual machines on the same isolated, host-only network:

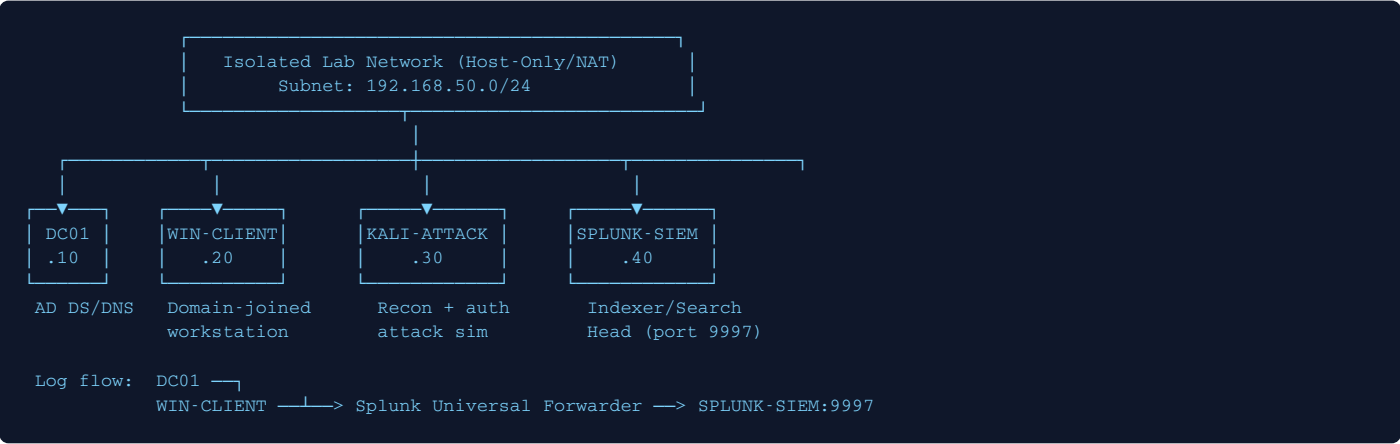
- DC01** (Windows Server 2025) — Active Directory Domain Services + DNS
- WIN-CLIENT** (Windows 10/11) — domain-joined end-user workstation
- KALI-ATTACK** (Kali Linux) — attacker machine for recon and credential-attack simulation
- SPLUNK-SIEM** (Ubuntu Server) — Splunk Enterprise indexer/search head

Lab Goals

- Stand up a Windows Server 2025 domain controller and promote a real AD forest
- Join a Windows client to the domain and authenticate as a domain user
- Deploy Splunk Enterprise and forward Windows Security/System/Application logs via Universal Forwarder
- Simulate recon and failed-authentication activity from Kali Linux
- Build detections: failed logons, account lockouts, privileged group changes, and Kerberos ticket requests

Architecture Diagram

Traffic flow and roles in this lab:



2. Prerequisites

Host Machine Requirements

Component	Minimum Requirement
RAM	16 GB (32 GB recommended for smooth four-VM operation)
Disk Space	250 GB free (SSD strongly recommended)
CPU	6+ cores with Intel VT-x / AMD-V enabled in BIOS
OS	Windows 10/11, macOS 12+, or Linux (Ubuntu/Debian)
Hypervisor	VMware Workstation Pro 17 or later (free for personal use)
Internet	Required for ISO downloads, Windows activation grace period, and Splunk installation

Software Downloads (All Free)

Item	Source
VMware Workstation Pro	vmware.com/products/workstation-pro.html

Windows Server 2025 ISO	Microsoft Evaluation Center (180-day eval)
Windows 10/11 ISO	Microsoft Software Download site
Kali Linux ISO/VM	kali.org/get-kali (prebuilt VMware image recommended)
Splunk Enterprise	splunk.com/download (free Developer license, 500 MB/day)
Splunk Universal Forwarder + Windows Add-on	splunk.com/download & Splunkbase

✔ **Good to Know**
VMware Workstation Pro has been free for personal use since May 2024. You will need to create a free Broadcom account to download it.

3. Lab Environment Setup — VMware Workstation Pro

3.1. Build an Isolated Lab Network

1. Open VMware Workstation Pro → **Edit → Virtual Network Editor** (run as Administrator).
2. Click **Add Network** and choose an unused VMnet (e.g., a new custom network).
3. Set the type to **NAT** — this isolates the lab from your home LAN while still allowing internet access for updates and tool downloads.
4. Set the subnet to `192.168.50.0` , mask `255.255.255.0` .
5. **Uncheck** "Use local DHCP service" — every host gets a static IP so domain name resolution stays reliable.
6. Click Apply, then assign every lab VM's network adapter to this VMnet.

Host	IP Address	Subnet Mask	Gateway	DNS
DC01	192.168.50.10	255.255.255.0	192.168.50.2	127.0.0.1 (self)
WIN-CLIENT	192.168.50.20	255.255.255.0	192.168.50.2	192.168.50.10
KALI-ATTACK	192.168.50.30	255.255.255.0	192.168.50.2	192.168.50.10
SPLUNK-SIEM	192.168.50.40	255.255.255.0	192.168.50.2	192.168.50.10

3.2. Create the Domain Controller VM (DC01)

1. Go to **File → New Virtual Machine** (or Ctrl+N). Select **Custom**.
2. Choose **Installer disc image file (iso)**, browse to the Windows Server 2025 ISO.
3. Guest OS: Windows, Version: Windows Server 2025. Click Next.

Virtual Machine Name	DC01
Disk Size	60 GB (store as single file)
Memory	4096 MB (4 GB)
Processors	2 cores
Network Adapter	Custom — lab VMnet (Section 3.1)

4. Click Finish. Power on the VM and install with the **Desktop Experience** option for easier GUI management.
5. After install, remove the ISO: VM > Settings > CD/DVD > disconnect.

3.3. Create the Windows Client VM (WIN-CLIENT)

Virtual Machine Name	WIN-CLIENT
Disk Size	60 GB (store as single file)
Memory	4096 MB (4 GB)
Processors	2 cores
Network Adapter	Custom — lab VMnet (Section 3.1)

Install Windows 10/11 normally; you'll rename and domain-join it in Section 5.

3.4. Create the Kali Linux VM (KALI-ATTACK)

Importing the prebuilt VMware image from kali.org saves time over installing from ISO.

Virtual Machine Name	KALI-ATTACK
Disk Size	40 GB
Memory	4096 MB (4 GB)
Processors	2 cores
Network Adapter	Custom — lab VMnet (Section 3.1)

3.5. Create the Splunk SIEM VM (SPLUNK-SIEM)

Virtual Machine Name	SPLUNK-SIEM
Guest OS	Ubuntu Server 22.04 LTS
Disk Size	60 GB

Memory	4096 MB (4 GB)
Processors	2 cores
Network Adapter	Custom — lab VMnet (Section 3.1)

During Ubuntu Server install, tick **Install OpenSSH server** so you can SSH in from your host for easier copy-paste of commands.

3.6. Take a Snapshot Before You Begin

VMware snapshots let you roll back instantly if something breaks.

1. With each VM powered on after its base OS install, go to **VM → Snapshot → Take Snapshot**.
2. Name it `Base Install — Clean State` and click Take Snapshot.

Repeat for all four VMs. You can always revert with VM > Snapshot > Revert to Snapshot.

⚠ **Important**

Keep the lab VMnet set to Host-only/NAT — never Bridged — for this lab. This protects your real home network from lab DHCP, DNS, and Kali traffic.

4. Windows Server 2025 — Domain Controller Configuration

4.1. Initial Configuration

1. Open Server Manager → Local Server, rename the computer to `DC01`, then reboot.
2. Set a static IP via Network Connections → IPv4 Properties:

IP address	192.168.50.10
Subnet mask	255.255.255.0
Gateway	192.168.50.2
Preferred DNS	127.0.0.1

4.2. Install Active Directory Domain Services

1. Server Manager → **Add Roles and Features**.
2. Select **Active Directory Domain Services** (DNS Server is auto-selected too) → Install.
3. Once complete, click the notification flag → **Promote this server to a domain controller**.
4. Choose **Add a new forest**. Root domain name, e.g. `corp.local`.
5. Set the Forest/Domain functional level to **Windows Server 2025**.
6. Set a DSRM (Directory Services Restore Mode) password and record it securely.
7. Accept the default NETBIOS name (`CORP`) and default paths, then click **Install**.

The server reboots automatically and is now your domain controller.

4.3. Create OUs, Users, and Groups

Open **Active Directory Users and Computers** (`dsa.msc`) and build out a basic OU structure:

```
corp.local
├── OU=Lab-Users
│   ├── jdoe (standard user)
│   └── asmith (standard user)
├── OU=Lab-Admins
│   └── admin.svc (delegated admin — not Domain Admin)
└── OU=Lab-Computers
    └── WIN-CLIENT lands here after domain join
```

Quick PowerShell to create a test OU and user (run on DC01 as Administrator):

```
New-ADOrganizationalUnit -Name "Lab-Users" -Path "DC=corp,DC=local"
New-ADUser -Name "Jane Doe" -SamAccountName "jdoe" `
  -UserPrincipalName "jdoe@corp.local" `
  -Path "OU=Lab-Users,DC=corp,DC=local" `
  -AccountPassword (ConvertTo-SecureString "ChangeMe!2025" -AsPlainText -Force) `
  -Enabled $true -ChangePasswordAtLogon $true
```

4.4. Configure an Account Lockout Policy

This policy gives you a real "defense triggers" scenario to test later from Kali, similar to rate-limiting in a WAF.

1. Open **Group Policy Management** (`gpmc.msc`) → Default Domain Policy → Edit.
2. Navigate to: Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies → Account Lockout Policy.
3. Set **Account lockout threshold**: 5 invalid attempts.
4. Set **Account lockout duration**: 15 minutes. Set **Reset counter after**: 15 minutes.

✓ Good to Know

A 180-day evaluation license is plenty for a home lab. You can re-arm the evaluation period or rebuild from a snapshot if you need more time.

5. DNS & Domain Join — Windows Client

5.1. Configure Static IP and DNS

On WIN-CLIENT, set a static IP that points DNS at the domain controller — this is critical for domain join to succeed:

IP address	192.168.50.20
Subnet mask	255.255.255.0
Gateway	192.168.50.2
Preferred DNS	192.168.50.10

Confirm name resolution works before joining:

```
nslookup corp.local
```

5.2. Join the Domain

1. System Properties → Computer Name → Change.
2. Select **Domain**, enter `corp.local`.
3. Authenticate with the domain administrator account when prompted.
4. Reboot the client.

5.3. Verify Domain Login

Log in with `CORP\jdoe` (or whichever user you created in Section 4.3) to confirm domain authentication works end to end.

⚠ Common Pitfall

If domain join fails with "cannot locate a domain controller," the client's DNS is almost always still pointed at the NAT gateway or a public resolver instead of 192.168.50.10. Fix the DNS setting and retry.

6. Kali Linux Attacker Configuration

6.1. Set a Static IP

```
sudo nmcli con mod "Wired connection 1" ipv4.addresses 192.168.50.30/24
sudo nmcli con mod "Wired connection 1" ipv4.gateway 192.168.50.2
sudo nmcli con mod "Wired connection 1" ipv4.dns "192.168.50.10"
sudo nmcli con mod "Wired connection 1" ipv4.method manual
sudo nmcli con up "Wired connection 1"
```

6.2. Update Tools

```
sudo apt update && sudo apt full-upgrade -y
```

6.3. Confirm Reachability to the Domain Controller

```
ping 192.168.50.10
nmap -sV -p 88,389,445,464,636,3268 192.168.50.10
```

You should see Kerberos (88), LDAP (389), SMB (445), and Global Catalog (3268) open — confirming Kali can reach the AD services you'll generate detectable activity against in Section 8.

Scope Note

This guide focuses on environment setup and detection, not deep exploitation walkthroughs. Once the lab is live, treat your DC like a real authorized test target — research each technique from a reputable source (MITRE ATT&CK, vendor docs, SANS) before running it.

7. Installing & Configuring Splunk SIEM

7.1. Install Splunk Enterprise

On SPLUNK-SIEM (Ubuntu Server):

```
wget -O splunk.deb "https://download.splunk.com/products/splunk/releases/<version>/linux/splunk-<version>-linux-amd64.deb"
sudo dpkg -i splunk.deb
sudo /opt/splunk/bin/splunk start --accept-license
```

Set an admin username/password at first start, then log in at:

```
http://192.168.50.40:8000
```

7.2. Enable Receiving From Forwarders

1. Settings → Forwarding and receiving → Configure receiving → New Receiving Port.
2. Add port `9997` and save.

7.3. Install the Splunk Add-on for Microsoft Windows

Download from Splunkbase, then upload via **Apps → Manage Apps → Install app from file**. This gives proper field extractions for Windows Event Log data (EventCode, Account_Name, Logon_Type, etc.).

7.4. Install Universal Forwarder on DC01 and WIN-CLIENT

```
msiexec.exe /i splunkforwarder.msi RECEIVING_INDEXER="192.168.50.40:9997" AGREETOLICENSE=Yes /quiet
```

7.5. Configure Log Inputs

On DC01 and WIN-CLIENT, edit `C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf`:

```
[WinEventLog://Security]
disabled = false
index = wineventlog

[WinEventLog://System]
disabled = false
index = wineventlog

[WinEventLog://Application]
disabled = false
index = wineventlog
```

```
net stop SplunkForwarder
net start SplunkForwarder
```

On the Splunk indexer, create the `wineventlog` index: Settings → Indexes → New Index.

7.6. Verify Ingestion

```
index=wineventlog host=DC01
```

You should see Security/System/Application events flowing in within a few minutes.

⚠ Save Your Credentials

Record the Splunk admin password immediately after first setup — there's no recovery prompt later if you forget it on a fresh install.

8. Attack Simulation & Detection from Kali Linux

8.1. Establish a Baseline

On WIN-CLIENT, log out and attempt a login with a deliberately wrong password 3 times, then check Splunk:

```
index=wineventlog EventCode=4625
```

4625 = failed logon. You should see the attempts with source workstation, account name, and failure reason.

8.2. Simulate Authentication Attempts From Kali

From KALI-ATTACK, attempt an SMB authentication against the DC with an intentionally invalid credential using a tool such as `netexec` (formerly crackmapexec):

```
netexec smb 192.168.50.10 -u jdoe -p 'WrongPassword123'
```

This generates failed-logon telemetry (EventCode 4625, and eventually 4740 if it crosses your lockout threshold from Section 4.4) that's visible back in Splunk — the same way a real credential-stuffing or password-spray attempt would surface.

8.3. Observe the Events in Splunk

1. Return to Splunk Web on SPLUNK-SIEM.
2. Run `index=wineventlog EventCode=4625 OR EventCode=4740`.
3. You should see the failed attempts and, once the lockout threshold is hit, an account-lockout event — including source IP, target account, and workstation name.

8.4. Kerberos Service Enumeration

From Kali, confirm Kerberos is reachable and observe the resulting authentication telemetry on the DC:

```
nmap -p 88 --script krb5-enum-users --script-args krb5-enum-users.realm='corp.local' 192.168.50.10
```

Cross-reference any resulting authentication activity against `EventCode=4768` (Kerberos TGT request) and `4769` (Kerberos service ticket request) in Splunk — these are the same event IDs used to detect real-world Kerberoasting activity in production environments.

9. Splunk — Advanced Configurations

9.1. Build a Real-Time Failed-Login Alert

1. In Splunk Web, run: `index=wineventlog EventCode=4625 | stats count by Account_Name, src`
2. Click **Save As → Alert**.
3. Set trigger condition: **Number of Results** > 5, time range **Real-time** or a 5-minute rolling window.
4. Add an action (e.g., log to triggered alerts, or email if you've configured an SMTP server).

9.2. Install Sysmon for Deeper Visibility

Installing **Sysmon** on DC01 and WIN-CLIENT (with a baseline config such as the SwiftOnSecurity ruleset) dramatically improves visibility — process creation, network connections, and registry changes all become searchable in Splunk.

```
Sysmon64.exe -accepteula -i sysmonconfig.xml
```

Add a matching stanza to `inputs.conf` on the forwarder:

```
[WinEventLog://Microsoft-Windows-Sysmon/Operational]
disabled = false
index = wineventlog
renderXml = true
```

9.3. Block Kali at the Network Level — Windows Firewall Rule

Create a rule on DC01 that blocks inbound traffic from the Kali IP, then re-test:

```
New-NetFirewallRule -DisplayName "Block-Kali-Lab" -Direction Inbound `
-RemoteAddress 192.168.50.30 -Action Block
```

From Kali, re-run the SMB authentication attempt from Section 8.2 — the connection should now be rejected, and you can observe the absence of new events in Splunk as confirmation the block is effective. Remove the rule when testing is complete:

```
Remove-NetFirewallRule -DisplayName "Block-Kali-Lab"
```

9.4. Add Splunk Security Essentials

Install the free **Splunk Security Essentials** app from Splunkbase for a library of MITRE ATT&CK-mapped detection searches you can enable against the data sources you've already onboarded.

Tip

Take a fresh VMware snapshot of each VM after completing Section 9 — you now have a fully wired detection pipeline, and it's worth preserving as a clean rollback point before experimenting further.

10. Conclusion & Further Exploration

Congratulations! You have built a fully functional Active Directory security lab demonstrating:

- VM provisioning and isolated networking in VMware Workstation Pro
- A real Windows Server 2025 Active Directory forest with users, OUs, and group policy
- Domain join and authentication from a Windows client
- End-to-end log forwarding into a Splunk SIEM via Universal Forwarder
- Detection of failed logons, account lockouts, and Kerberos ticket activity generated from Kali Linux

Ideas for Further Exploration

Extension	Description
BloodHound	Map AD attack paths and trust relationships visually
Atomic Red Team	Run small, mapped MITRE ATT&CK test cases and validate each one in Splunk
Additional domain-joined hosts	Add a second client or member server to widen the detection surface
Wazuh or Elastic comparison	Stand up an alternate SIEM and compare detection coverage
Certificate Services (AD CS)	Add a CA role and explore certificate-based attack/detection scenarios
Tiered Admin Model	Rebuild the lab using a tiered privileged-access model and measure the detection difference
Container Lab	Replace the Splunk VM with a Docker-based Splunk instance for a lighter-weight setup

11. Troubleshooting Reference

Problem	Solution
Client can't join the domain	DNS must point at the DC, not the gateway or a public resolver. Set client DNS to 192.168.50.10.
<code>nslookup corp.local</code> fails	Confirm the DNS Server role is running on DC01: <code>Get-Service DNS</code> . Check Windows Firewall allows TCP/UDP 53.
Splunk shows no data	Test forwarder-to-indexer connectivity: <code>Test-NetConnection 192.168.50.40 -Port 9997</code>
Forwarder service won't start	Re-run the installer with the correct indexer address, or edit <code>outputs.conf</code> under <code>etc\system\local</code> .
VMs can't reach the internet	Confirm the VMnet type is set to NAT, not Host-only, in Virtual Network Editor.
Account lockout never triggers	Confirm the Default Domain Policy applied: run <code>gpupdate /force</code> on DC01 and re-check the lockout threshold value.
Kerberos events not appearing	Confirm Advanced Audit Policy has "Audit Kerberos Authentication Service" and "Audit Kerberos Service Ticket Operations" enabled under Account Logon.
Sysmon events missing from Splunk	Verify the channel name in <code>inputs.conf</code> exactly matches <code>Microsoft-Windows-Sysmon/Operational</code> .

Useful Commands Reference

```
# Check AD DS / DNS service health on DC01:
Get-Service NTDS,DNS,Netlogon,Kdc

# Force a Group Policy refresh:
gpupdate /force

# Check Splunk forwarder status on Windows:
"C:\Program Files\SplunkUniversalForwarder\bin\splunk.exe" status

# Check Splunk indexer is listening:
sudo ss -tlnp | grep 9997

# Quick reachability test from Kali:
nmap -sV -p 88,389,445,464,636,3268 192.168.50.10
```

Keep your home lab isolated from production networks · Regularly rotate the lab's snapshots · Good luck!